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Experiment 5

Input

2 3
aaaaa
acacaca
aabaa
ccacacc
caaac

514C

O/P

Yes
No
No

Watto the owner of a spare parts store, has recently got an order for the mechanism that can process strings in a certain way. Initially the memory of the mechanism is filled with n strings. Then the mechanism should be able to process queries of the following type: "Given string s, determine if the memory of the mechanism contains string t that consists of the same number of characters as s and differs from s in exactly at one position."

Watto has already compiled the mechanism, all that's left is to write a program for it and check it on the data consisting of n initial lines and m queries. He decided to entrust this job to you.

⇒ Approach :

1) Store all memory strings in a set

2. For each query string:
- Try changing each position
- Replace it with other two possible ~~to~~ characters.
- Check if modified string exists in set.

3. Because:
- only 3 characters → at each index only 2 replacements possible
- If length = L → total checks = 2L
- Total operations = O(total length)

⇒ <u>Code:</u>

```
# include <bits/stdc++.h>
using namespace std;

int main () {
    int n, m;
    cin >> n >> m;

    unordered_set< string > st;

    for (int i=0 ; i<n ; i++) {
        string s;
        cin >> s;
        st.insert (s);
    }
    while (m--) {
        string s;
        cin >> s;

        bool found = false;
```

```
for (int i=0; i< s.length() && !found; i++){
    char original = s[i]
    for (char ch = 'a'; ch <= 'c'; ch++){
        if (ch == original) continue;
        s[i] = ch;

        if (st.count(s)) {
            found = true
            break;
        }
    }
    s[i] = original;
}

if(found)
    cout << "YES\n";
else
    cout << "NO\n";
}
return 0;
}
```

O/P:

YES
No
No.